AI-Assisted Software Development

Module 2: Design and Architecture with AI Tools

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Overview

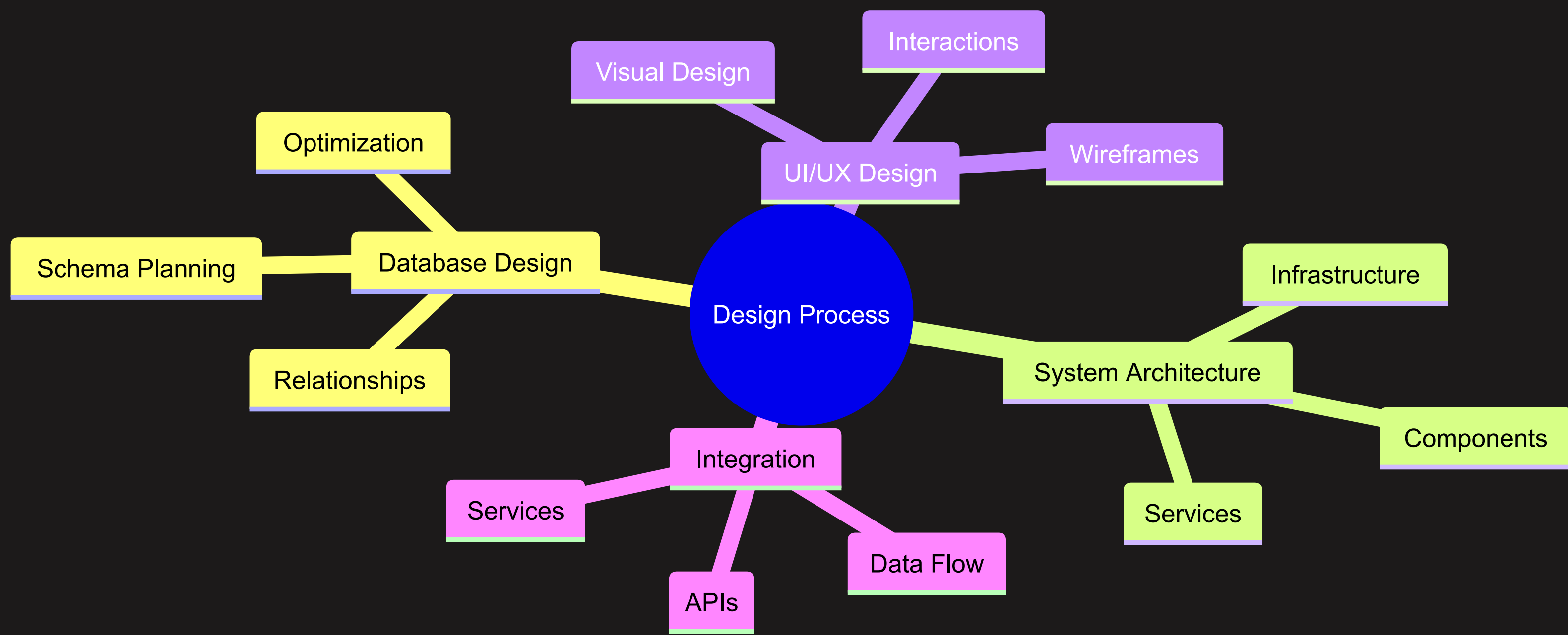
What we'll cover:

- Database Design
- System Architecture
- UI/UX Design
- Design Systems
- Technical Specifications

Tools we'll use:

- AI for architecture decisions
- V0 for UI generation
- Cursor for documentation
- AI for design validation

The Design Process



Database Design with AI

Start with your PRD and ask AI to:

1. Identify Data Entities

```
Core Entities:
- Users
- Projects
- Tasks
- Teams
- Comments
```

2. Define Relationships

```
Relationships:
- User belongs_to many Teams
- Team has_many Projects
- Project has_many Tasks
- Task has_many Comments
```

Database Schema Generation

Example AI conversation:

Input to AI:

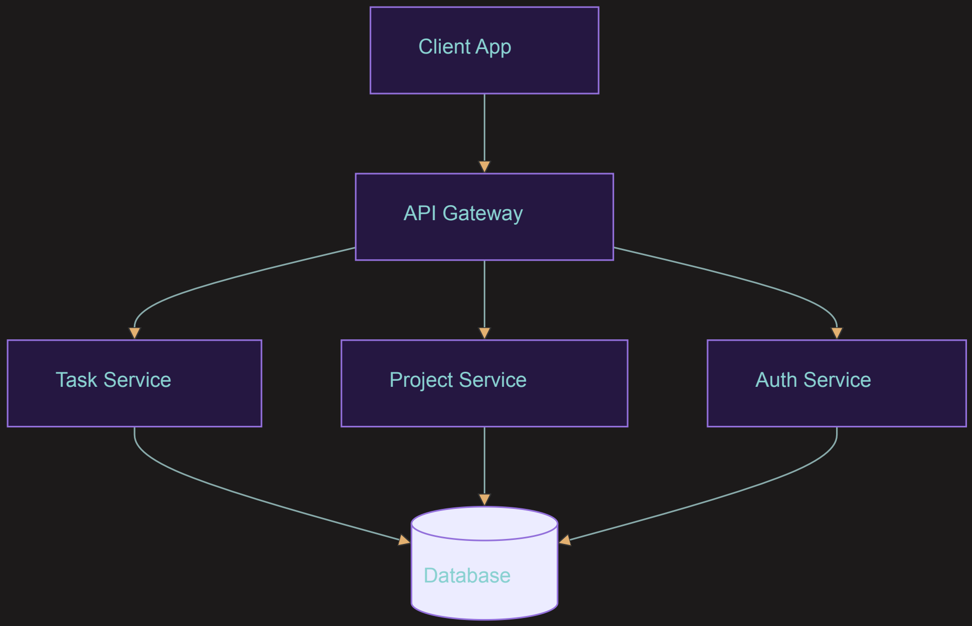
```
Generate a database schema for a
task management app with:
- User authentication
- Team collaboration
- Project organization
- Task dependencies
```

AI helps with:

- Table definitions
- Field types
- Relationships
- Indexes
- Constraints

System Architecture Planning

Use AI to design system components:



Break down into:

- Services
- Components
- Dependencies
- Data flow

UI/UX Design with V0

V0 workflow:

1. Start with Requirements

   ```
   Design a task management app:
   - Clean, modern interface
   - Mobile-responsive
   - Team collaboration focus
   ```

2. Add Inspiration
   - Reference similar apps
   - Include screenshots
   - Specify design preferences

3. Iterate Design
   - Refine through prompts
   - Focus on user experience
   - Maintain consistency

UI Design Best Practices with V0

1. Clear Initial Prompts

```
Create a modern task dashboard with:
- Left sidebar navigation
- Card-based task view
- Status indicators
- Team member avatars
```

2. Iterative Refinement

```
Adjust the design to:
- Use more whitespace
- Increase contrast
- Add hover states
- Improve hierarchy
```

Design System Creation

Use AI to help define:

1. Visual Elements
   - Colors
   - Typography
   - Spacing
   - Icons
2. Components
   - Buttons
   - Forms
   - Cards
   - Navigation
3. Patterns
   - Layouts
   - Interactions
   - Animations
   - States
4. Documentation
   - Usage guidelines
   - Examples
   - Best practices

Technical Specification Development

Convert designs to technical specs:

```
## Component: TaskCard

Properties:
- title: string
- description: string
- status: enum
- assignee: User
- dueDate: DateTime

Behaviors:
- Drag and drop
- Status toggle
- Quick edit
- Comment thread
```

Integration Planning

Plan how components work together:

1. API Design

```
POST /api/tasks
{
  "title": string,
  "project_id": uuid,
  "assignee_id": uuid,
  "due_date": datetime
}
```

2. State Management

```
{
  tasks: {
    byId: {},
    allIds: [],
    loading: boolean,
    error: string | null
  }
}
```

Design Validation

Use AI to validate designs for:

1. Usability
    - Navigation flow
    - User interactions
    - Accessibility
    - Mobile responsiveness
2. Technical Feasibility
    - Performance impact
    - Implementation complexity
    - Resource requirements
    - Scalability concerns

Documentation in Cursor

Maintain design documentation:

```
designs/
├── ui/
│   ├── components/
│   ├── layouts/
│   └── styles/
├── architecture/
│   ├── services/
│   └── infrastructure/
└── database/
    ├── schema/
    └── migrations/
```

Keep everything organized for AI context.

Best Practices

1. Design Process
    - Start with user needs
    - Design iteratively
    - Validate early
    - Document decisions
2. Tool Usage
    - Provide clear context
    - Use reference images
    - Iterate designs
    - Maintain consistency
3. Documentation
    - Keep organized
    - Update regularly
    - Link related files
    - Version control

Common Pitfalls

1. Design Phase
   - ❌ Skipping user research
   - ❌ Ignoring mobile views
   - ✅ User-centered design
   - ✅ Responsive first
2. Technical Planning
   - ❌ Over-engineering
   - ❌ Ignoring scalability
   - ✅ Start simple
   - ✅ Plan for growth

Recap

- Start with database design
- Plan system architecture
- Use V0 for UI design
- Create comprehensive design system
- Develop technical specifications
- Plan integration points
- Validate designs
- Maintain documentation

Next Steps

- Review designs with stakeholders
- Begin development setup
- Create component library
- Start implementation